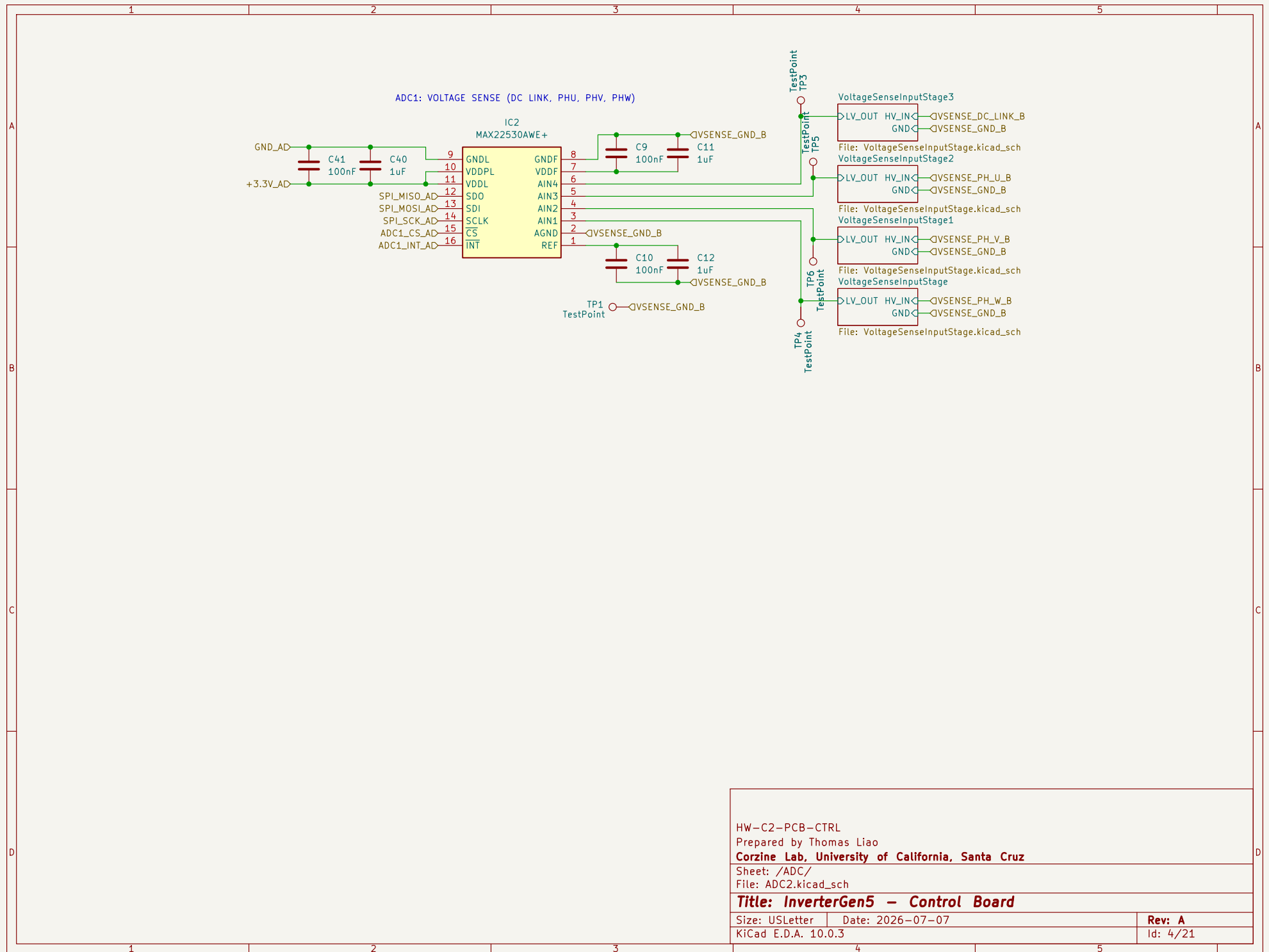


NOTE: WHEN NOT IN FIBER MULTIDRIVE MULTIPHASE OR LOAD SHARING OPERATION,
DISABLE TIM1 SYNC PIN (TIM1 ETR)



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Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/

File: ADC2.kicad_sch

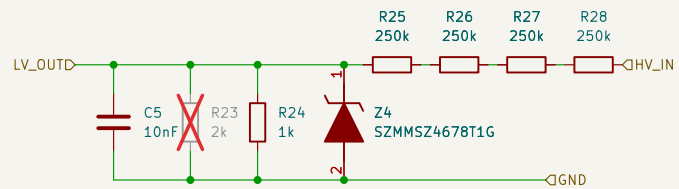
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 4/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage/

File: VoltageSenseInputStage.kicad_sch

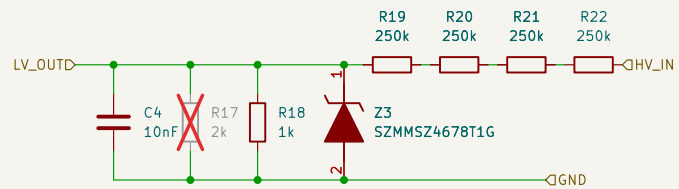
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

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Id: 2/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage1/

File: VoltageSenseInputStage.kicad_sch

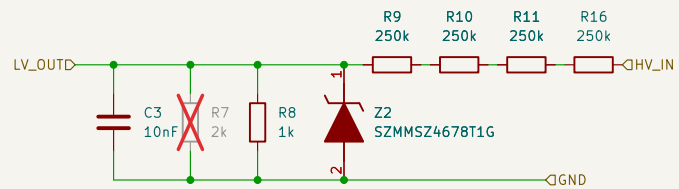
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

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Id: 3/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage2/

File: VoltageSenseInputStage.kicad_sch

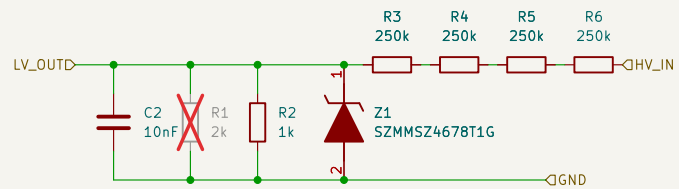
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

KiCad E.D.A. 10.0.3

Rev: A

Id: 5/21



ZENER CLAMPS ADC INPUT AT 1.8V
ADC MAX INPUT IS 1.8V ON SENSE PINS

HW-C2-PCB-CTRL

Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /ADC/VoltageSenseInputStage3/

File: VoltageSenseInputStage.kicad_sch

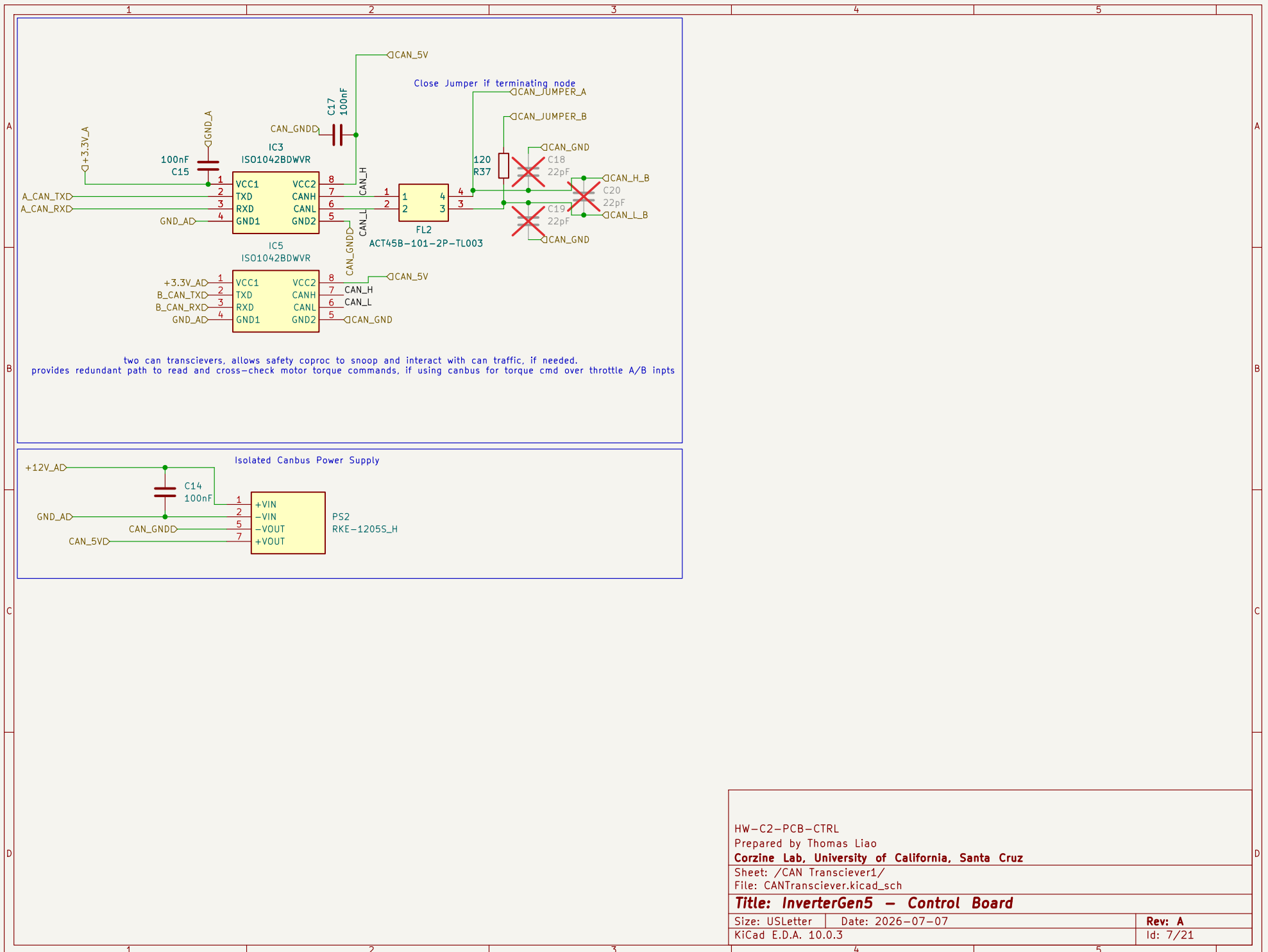
Title: InverterGen5 - Control Board

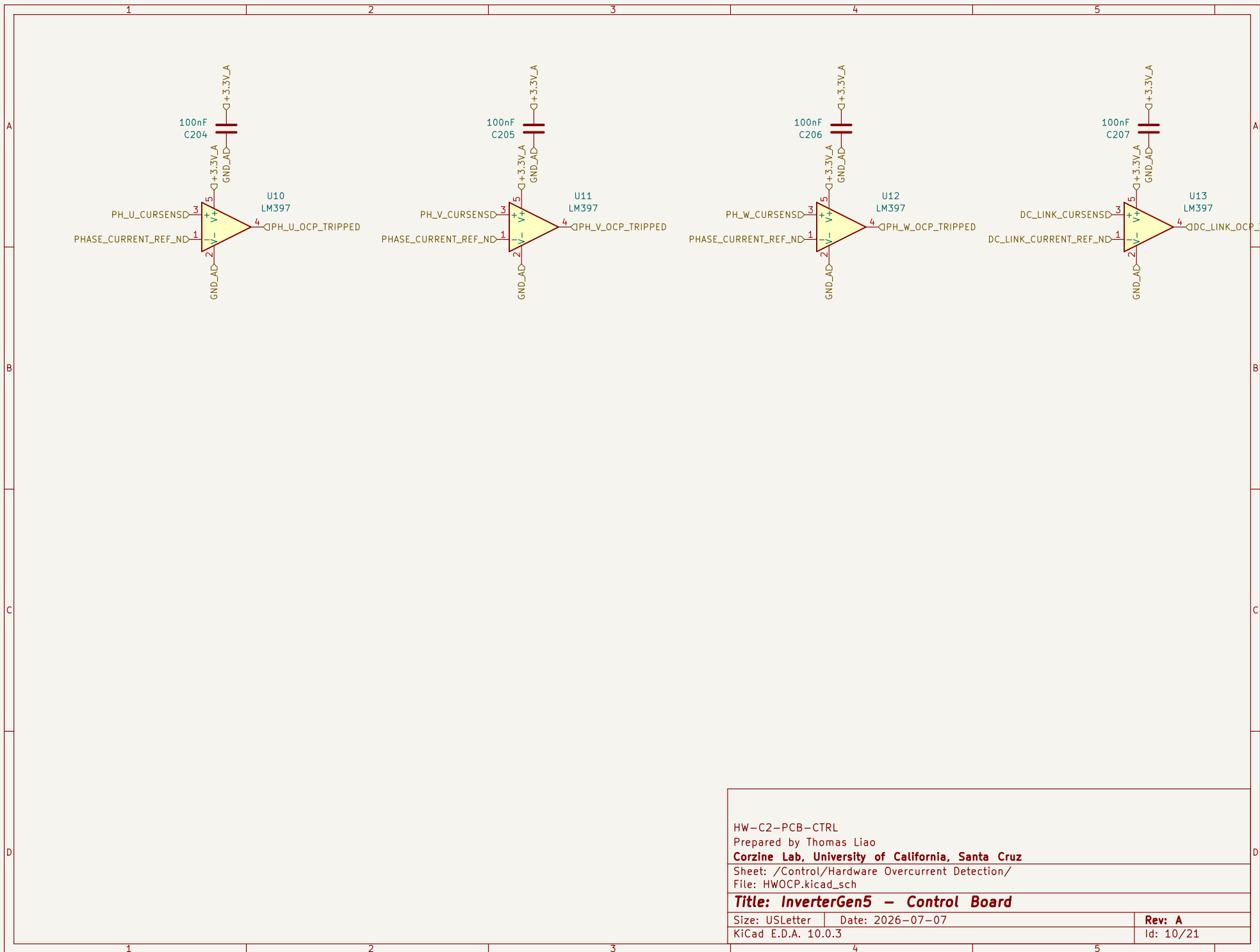
Size: USLetter Date: 2026-07-07

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Prepared by Thomas Liao

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Sheet: /Control/Hardware Overcurrent Detection/

File: HWOCP.kicad_sch

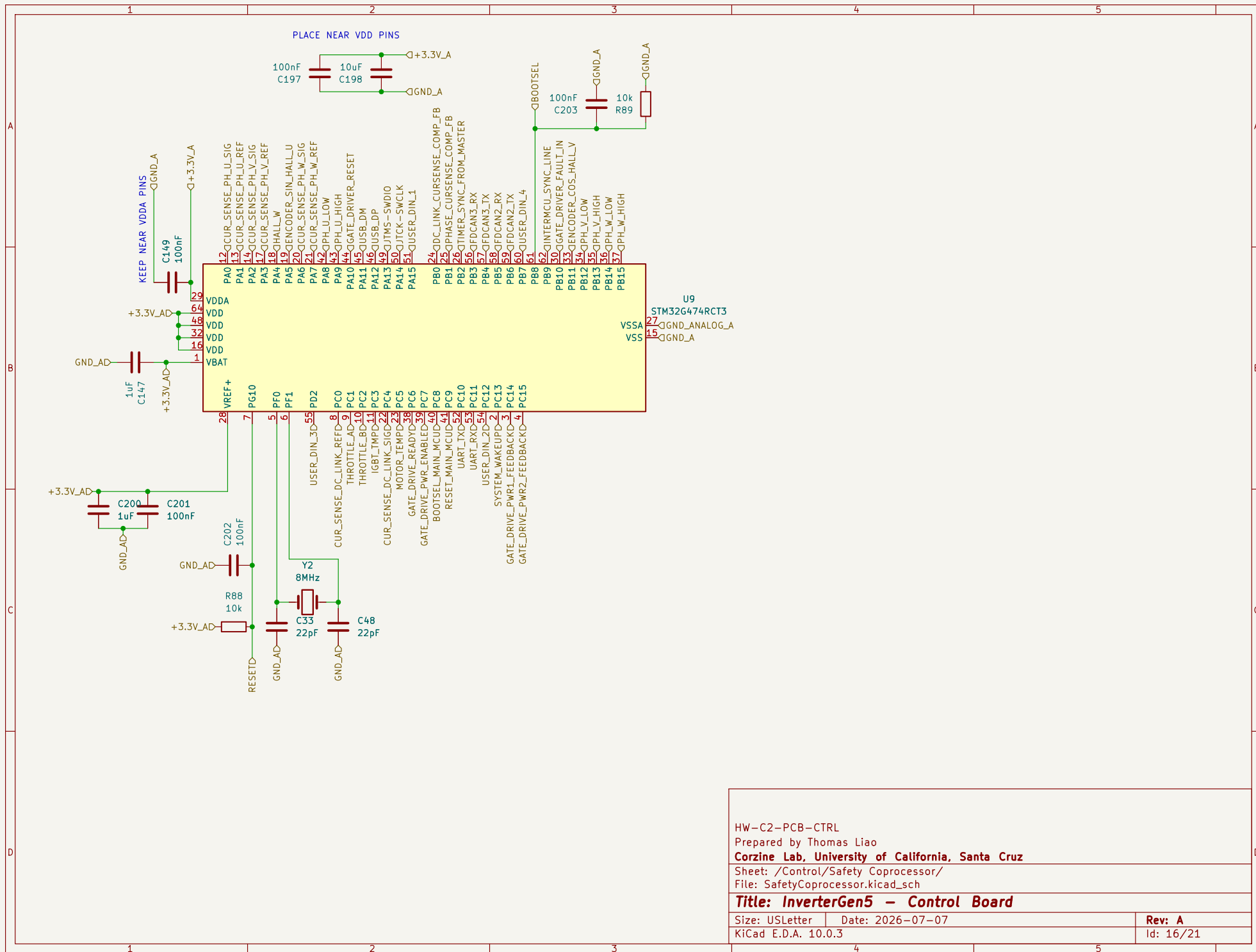
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

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Id: 10/21



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Prepared by Thomas Liao

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Sheet: /Control/Safety Coprocessor/

File: SafetyCoprocessor.kicad_sch

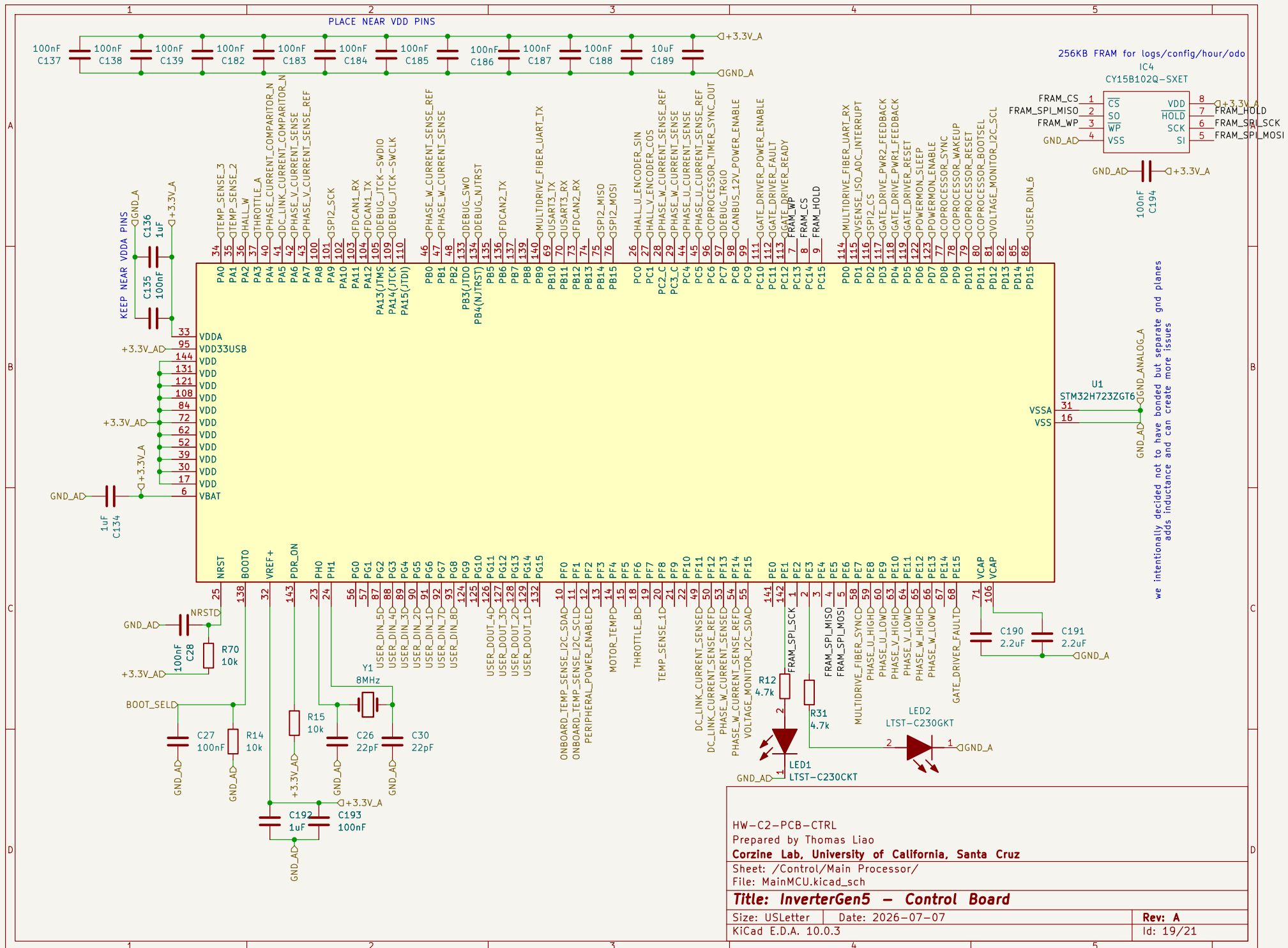
Title: InverterGen5 - Control Board

Size: USLetter Date: 2026-07-07

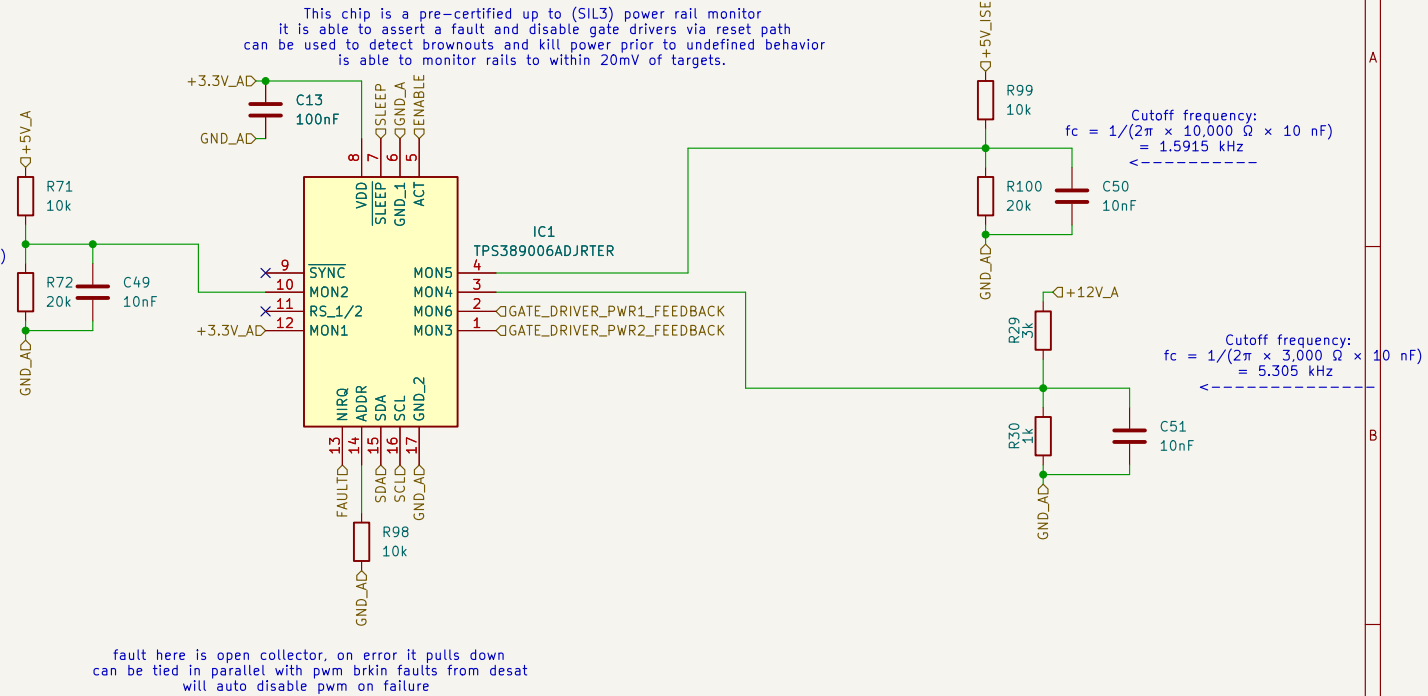
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$$\text{Cutoff frequency: } f_c = 1/(2\pi \times 10,000 \, \Omega \times 10 \, \text{nF}) = 1.5915 \, \text{kHz}$$



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Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Control/Power Supply Monitor/

File: PowerSupplyMonitor.kicad_sch

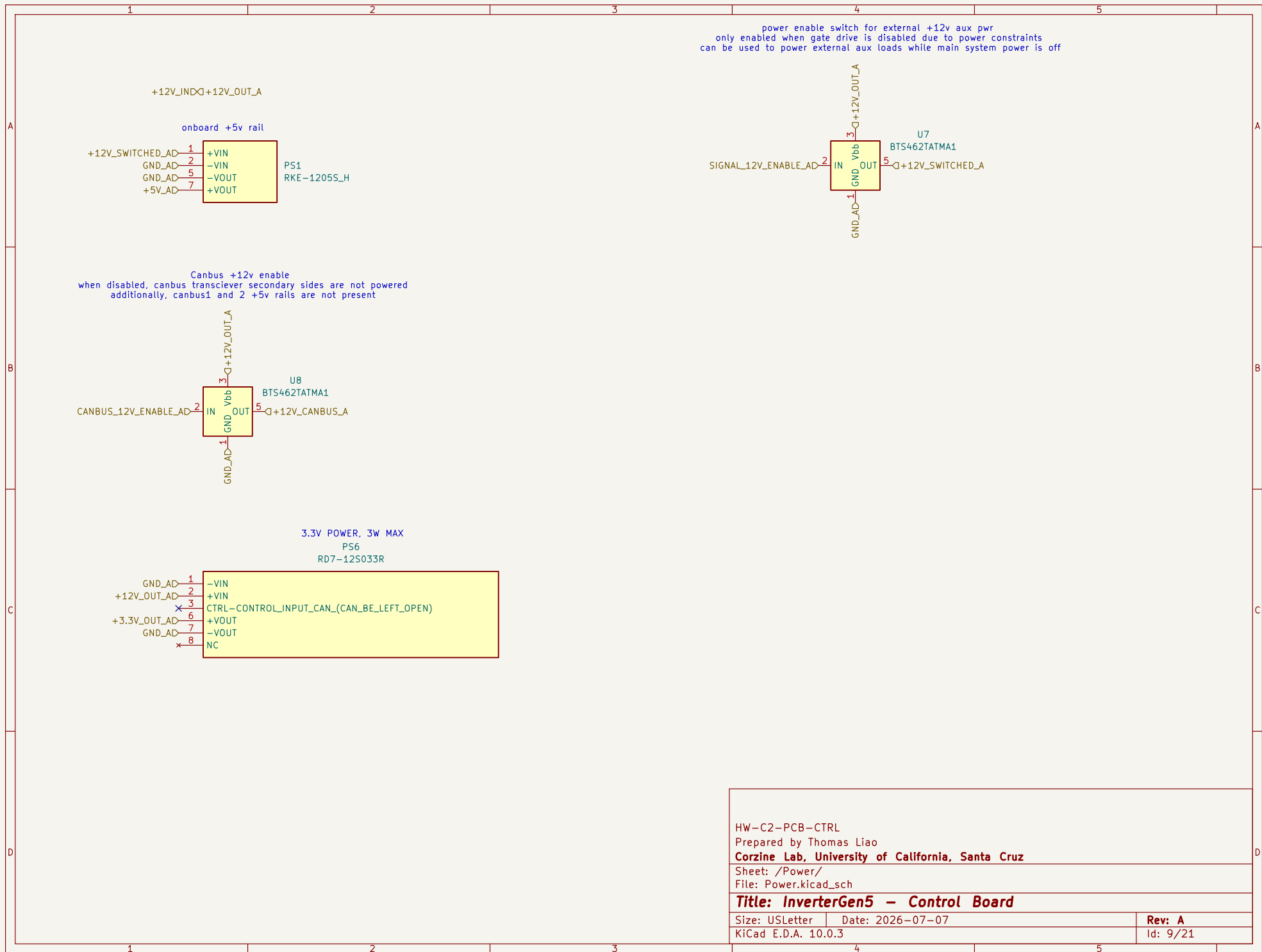
Title: InverterGen5 - Control Board

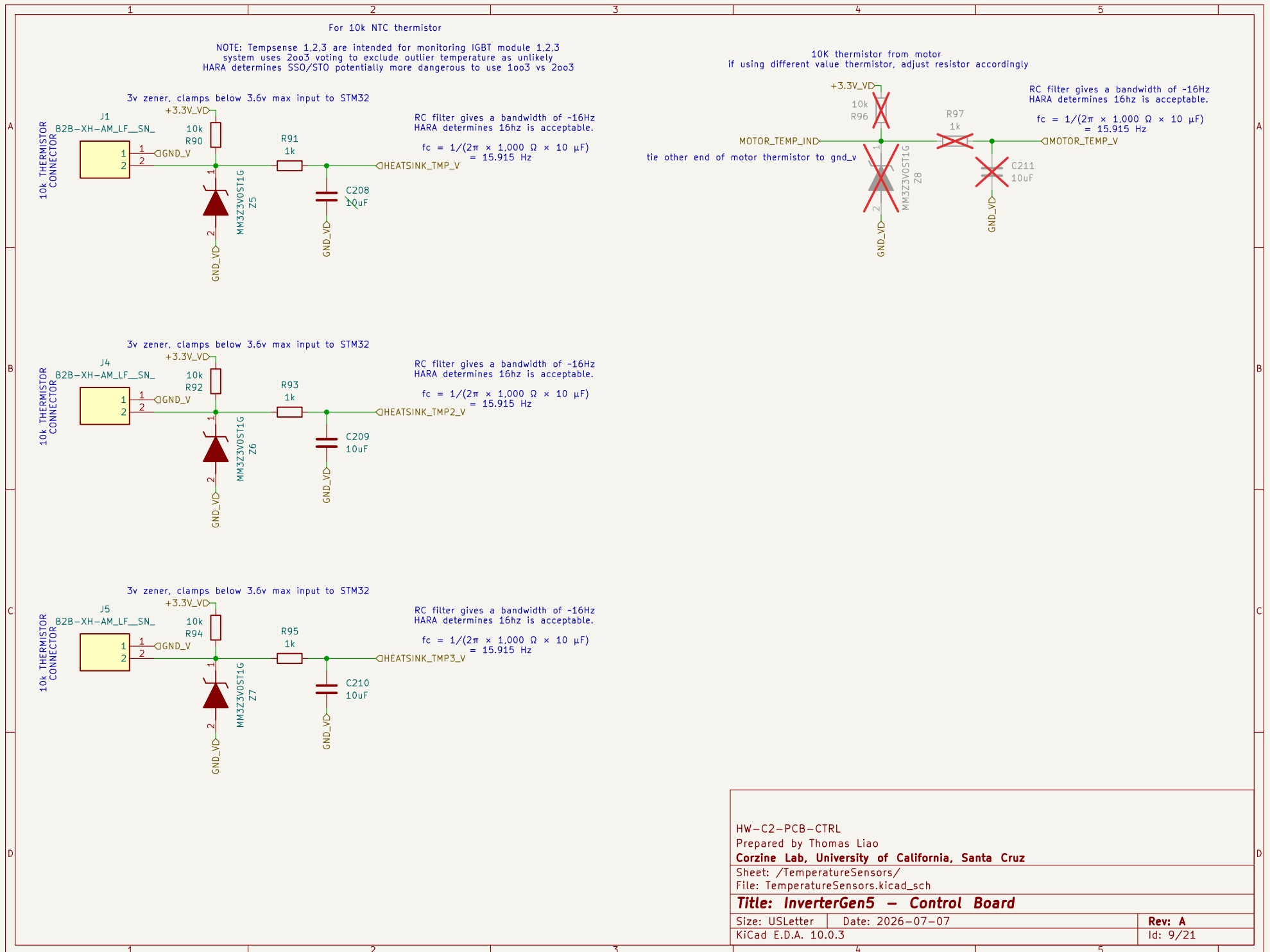
Size: USLetter Date: 2026-07-07

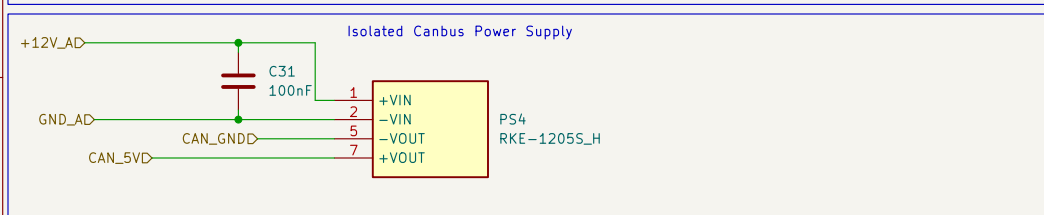
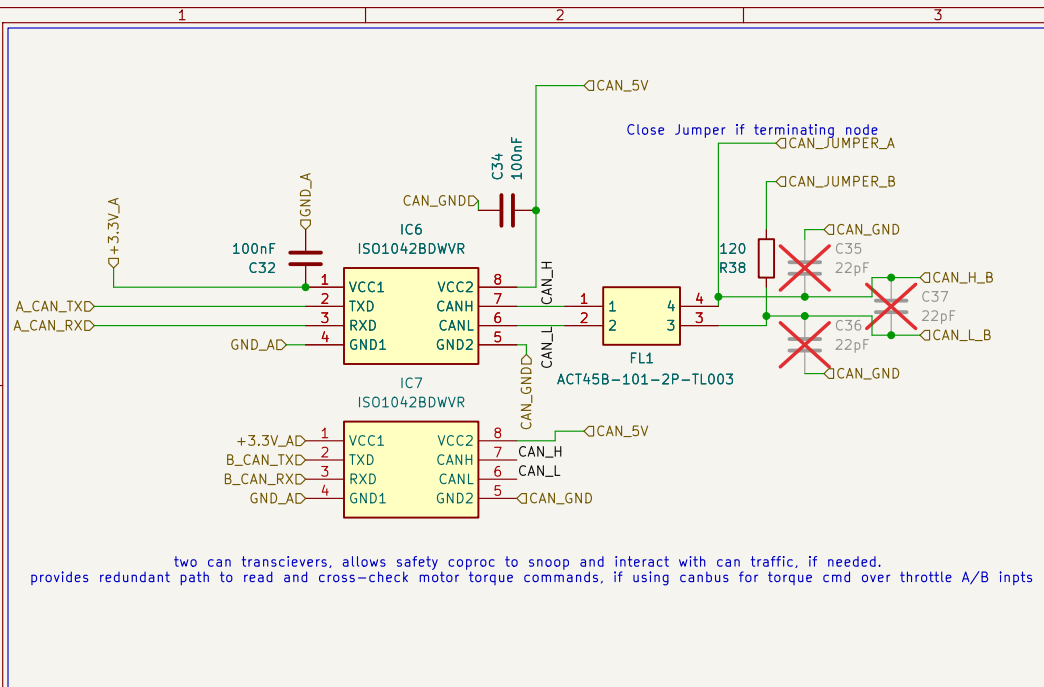
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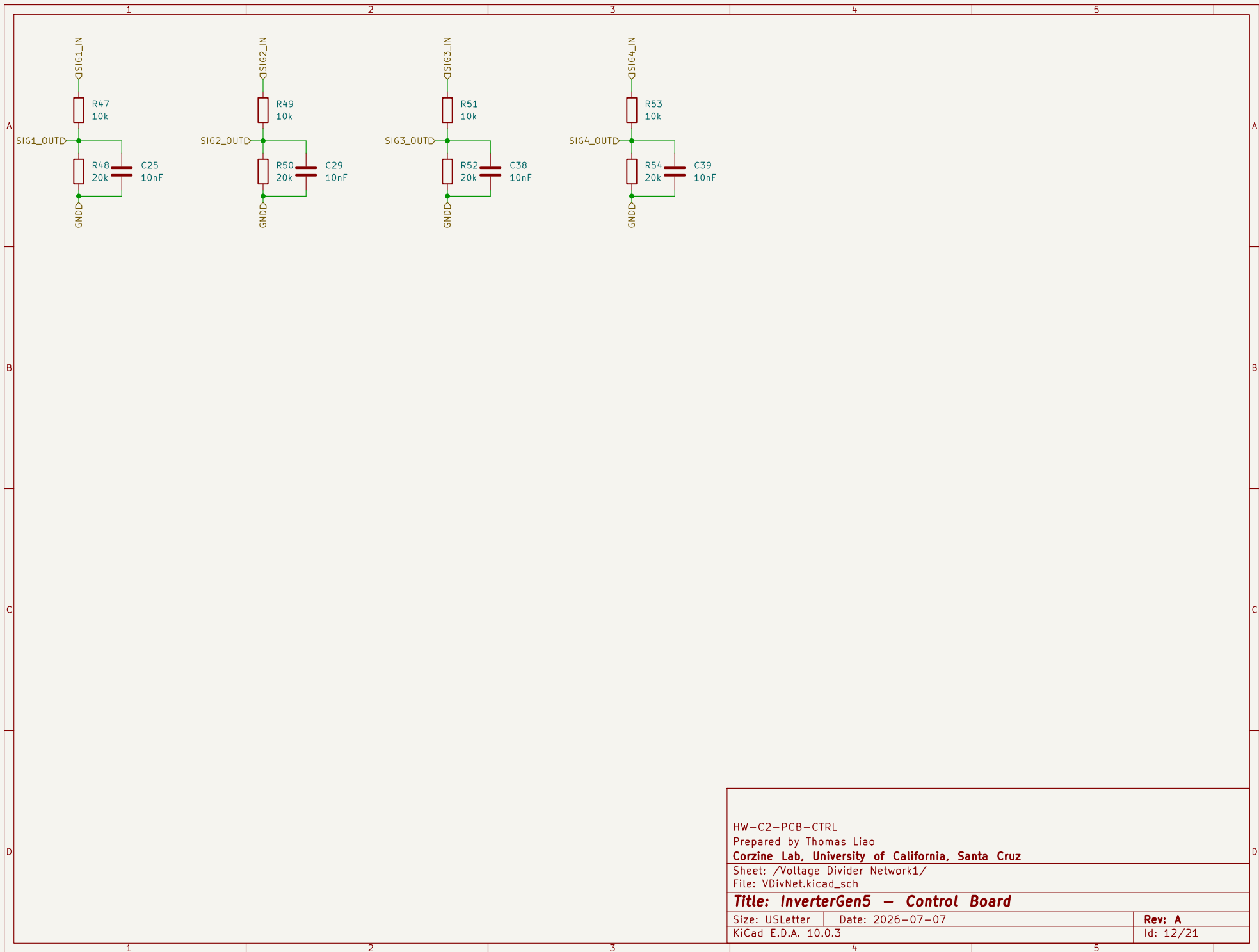
Rev: A

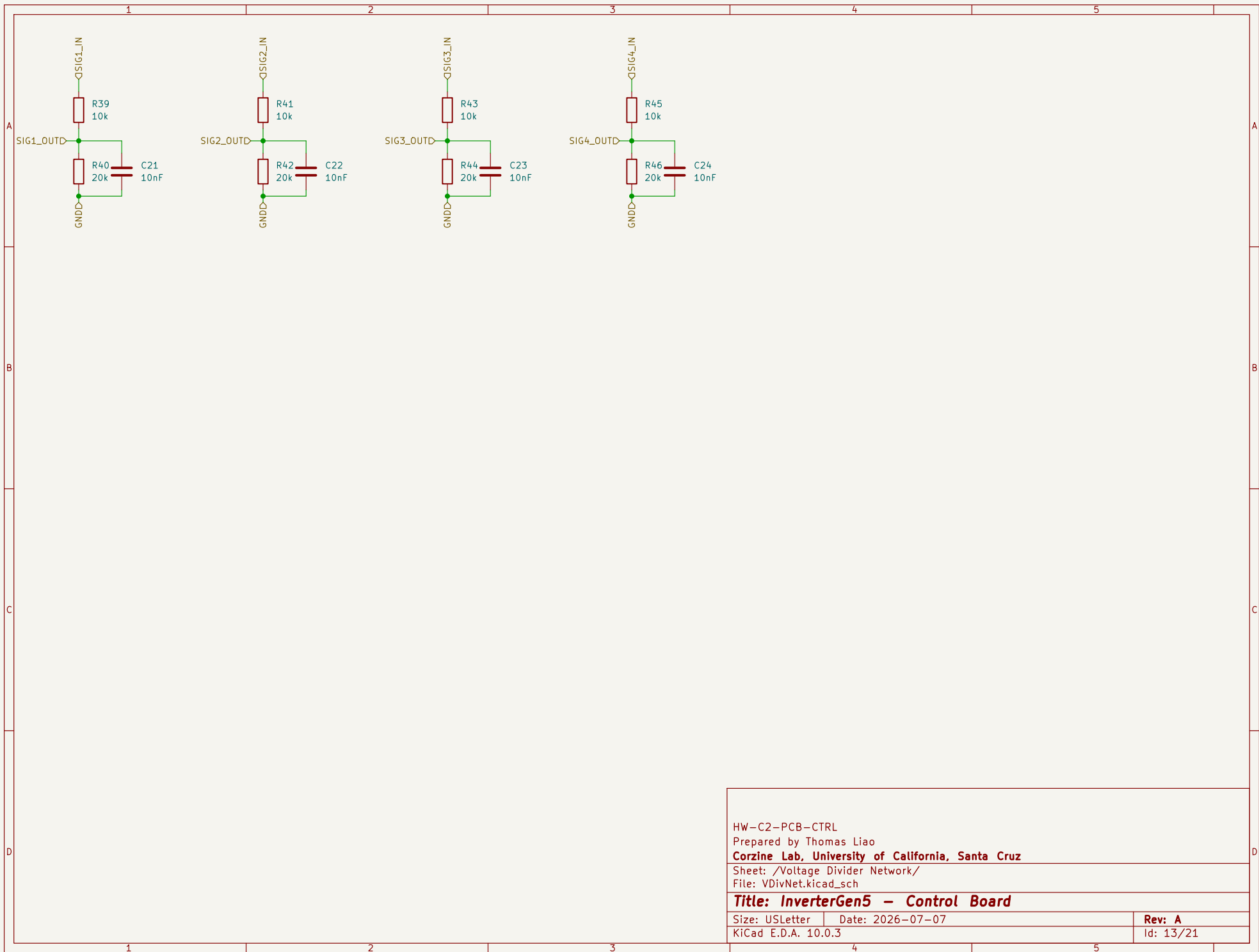
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Sheet: /Voltage Divider Network/

File: VDivNet.kicad_sch

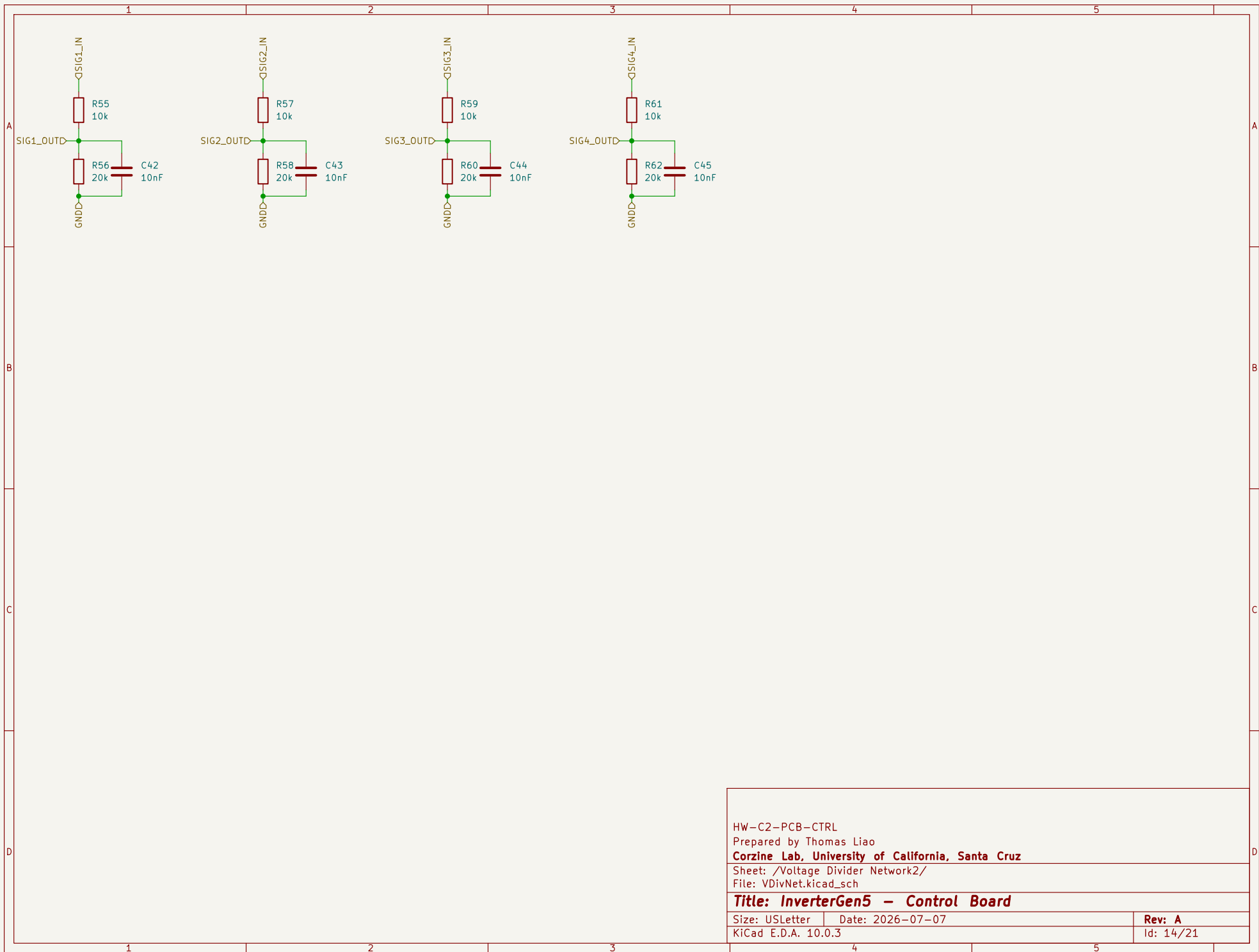
Title: InverterGen5 - Control Board

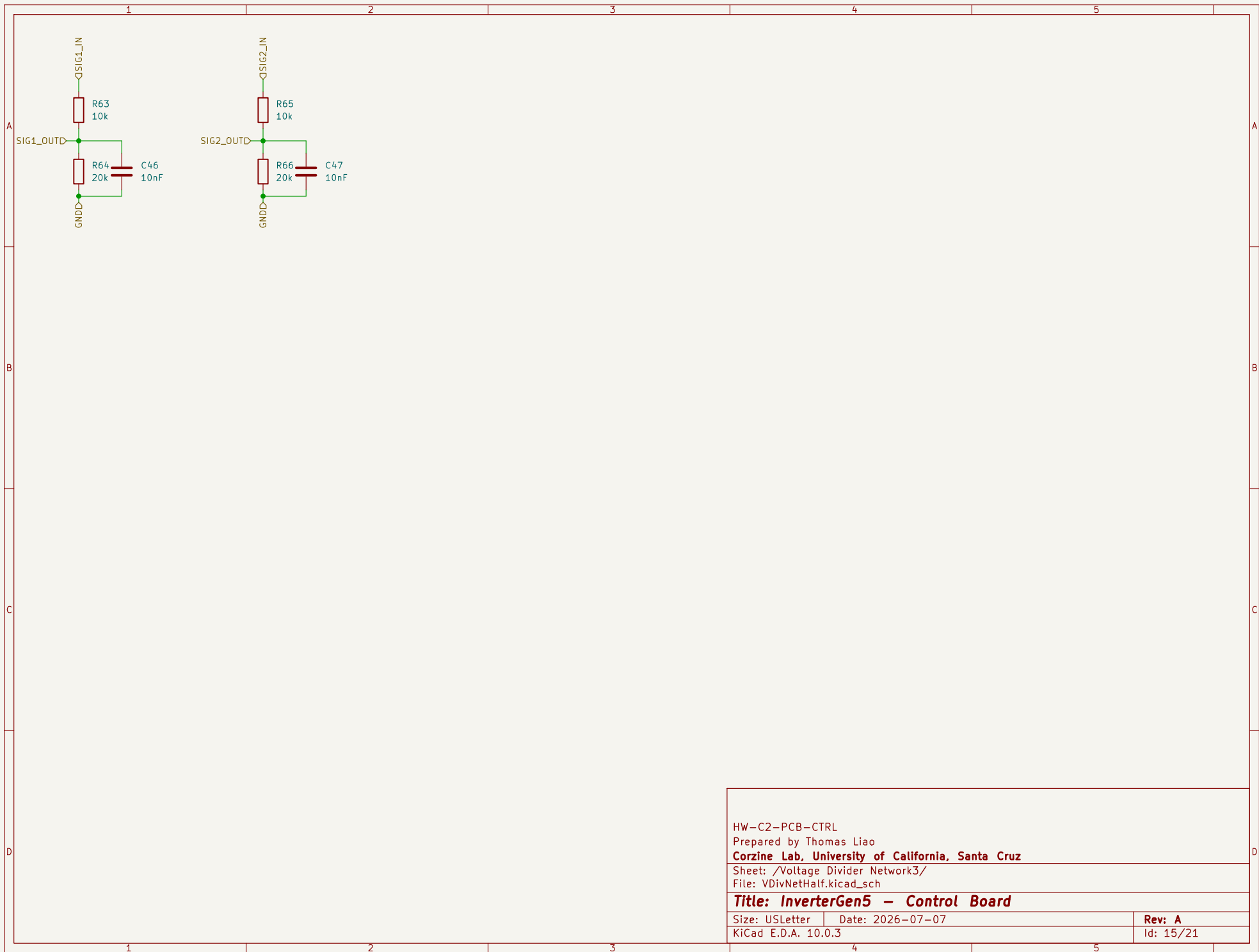
Size: USLetter Date: 2026-07-07

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Prepared by Thomas Liao

Corzine Lab, University of California, Santa Cruz

Sheet: /Voltage Divider Network3/

File: VDivNetHalf.kicad_sch

Title: InverterGen5 - Control Board

Size: USLetter

Date: 2026-07-07

Rev: A

KiCad E.D.A. 10.0.3

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